Computers & Security 154 (2025) 104403

Contents lists available at ScienceDirect

Computers & Security

journal homepage: www.elsevier.com/locate/cose

RADIANT: Reactive Autoencoder Defense for Industrial Adversarial Network Threats

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ARTICLE INFO

Keywords:
Network Intrusion Detection
Adversarial robustness
Reactive defense
Machine learning
Industrial Control Systems

ABSTRACT

Machine learning-based Intrusion Detection Systems (IDS) have significantly enhanced operational efficiency in Industrial Control Systems (ICS), but they face a growing threat from Adversarial Machine Learning (AML) attacks. These attacks exploit vulnerabilities in IDS, leading to delayed threat detection, infrastructure compromise, financial losses, and service disruptions. Traditional approaches, such as adversarial retraining, are not only resource-intensive but also suffer from limited generalization, as they rely on training models with specific adversarial samples. Given the constantly evolving nature of adversarial attacks, it is impractical to train on all possible attack variations, leaving systems vulnerable to new and unforeseen threats. To address these limitations, this paper introduces Reactive Autoencoder Defense for Industrial Adversarial Network Threats (RADIANT), a novel IDS that mitigates adversarial threats without relying on retraining. By reconstructing input data and analyzing three distinct reconstruction errors, RADIANT effectively reduces the impact of adversarial perturbations. To evaluate RADIANT's performance, we used a comprehensive assessment framework that compared it against state-of-the-art defenses and undefended baseline classifiers on real-world ICS data. The evaluation included advanced adversarial attacks, such as HopSkipJump and Zeroth-Order Optimization (ZOO), conducted under gray-box conditions. During ZOO attacks, RADIANT achieved an F1 score of 85.9%, significantly outperforming the baseline classifier's 17.1% and demonstrating its robustness against adversarial threats. Similarly, under HopSkipJump attacks, RADIANT maintained a strong F1 score of 91.4%, far exceeding the baseline's 20.5%. Additionally, when compared to state-of-the-art proactive defenses based on adversarial training, RADIANT consistently delivered a superior balance of precision, recall, and overall robustness, all without the need for adversarial retraining. These results highlight RADIANT's practicality and effectiveness, offering reliable protection for ICS while addressing the increasing sophistication of AML attacks.

1. Introduction

Industrial Control Systems (ICS) are integral components of Critical Infrastructure, encompassing vital sectors like manufacturing, power grids, water treatment, ports, oil refineries, and healthcare facilities (Bhamare et al., 2020; Koay et al., 2023; Pöyhönen, 2023; Mehdi et al., 2024). In the past, these ICS networks operated in relative isolation, shielded from cyber threats by maintaining no direct connections to the internet. However, the growing demand for connectivity, and remote monitoring has exposed ICS to various security vulnerabilities (Kravchik and Shabtai, 2018). This shift has made ICS systems a prime target for cyberattacks, raising concerns due to their pivotal role in controlling physical operations and the potential far-reaching consequences of breaches. Consequently, addressing security issues in ICS networks has become a global imperative, emphasizing the need for robust, secure, and efficient measures to detect and defend against cyber threats in these critical systems (Ashibani and Mahmoud, 2017).

Machine Learning-based IDSs play a crucial role in enhancing the security and robustness of ICS by carefully scrutinizing network and sensor data to uncover potential threats and anomalies that might affect ICS operations. They provide proactive defense capabilities against an ever-evolving landscape of cyber threats that target critical infrastructure. By integrating advanced Machine Learning (ML) algorithms, these IDSs excel at detecting emerging attack vectors that are often disregarded by traditional rule-based systems. Consequently, there has been a substantial increase in the adoption and incorporation of ML into IDSs (Farrukh et al., 2023, 2022; Trivedi et al., 2023; Kravchik and Shabtai, 2018; Teixeira et al., 2018). However, the adoption of such systems introduces an additional dimension of risk: the vulnerability of the trained models to attacks, commonly known as Adversarial Machine Learning (AML) (Koay et al., 2023; de Azambuja et al., 2023). The objective here is to exploit the model's "blind spots" between

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https://doi.org/10.1016/j.cose.2025.104403
Received 1 December 2024; Received in revised form 7 February 2025; Accepted 27 February 2025
Available online 7 March 2025
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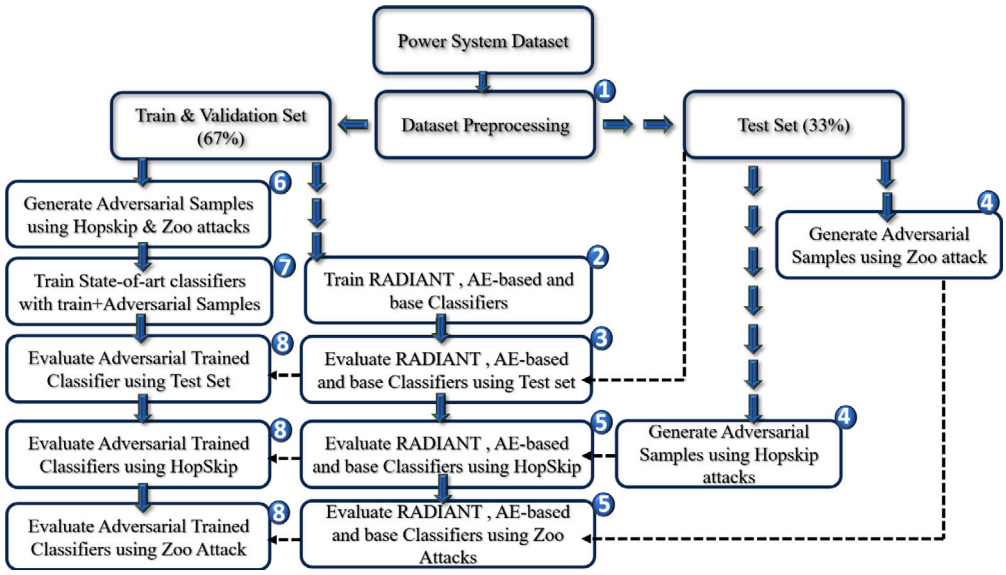

Fig. 1. Methodological framework: A visual representation of the research methodology, comprising dataset preprocessing, classifier training, evaluation, adversarial sample generation, and comparative analysis of adversarial robustness with state-of-the art and baseline classifiers.

the data points it encountered during training. This is achieved by introducing slight perturbations to unseen data points, causing the model to traverse a decision boundary and misclassify the data. This strategy, aligned with the boundary tilting hypothesis, targets the model's integrity, allowing adversaries to disguise malicious network traffic as benign (Tanay and Griffin, 2016). By exploiting these blind spots, adversaries can remain undetected by IDS, successfully achieving their malicious objectives.

In the context of ICS, adversarial attacks can manipulate data from actuators and devices by introducing subtle perturbations, allowing malicious data to bypass ML-based IDSs. This manipulation can lead to delayed attack detection, temporary infrastructure failures, data leakage, financial losses, and even threats to human life (Wei et al., 2018). Real-world demonstrations of these attacks have highlighted their significant potential for disruption and harm (Oprea and Vassilev, 2023).

To address the evolving threat of AML attacks, it is crucial that ML-based IDSs undergo thorough evaluation across a range of scenarios. This evaluation should account for the capabilities and objectives of potential attackers, as well as the design and assessment of attack methods that exploit vulnerabilities in machine learning during the development phase. While several proactive defense measures have been proposed in the past, many rely on adversarial retraining, which reduces their effectiveness when confronted with new or unseen adversarial attacks. To the best of our knowledge, this study represents the first investigation within the domain of ICS focused on developing a reactive defense mechanism against AML attacks. This defense is designed to identify adversarial attacks without requiring prior training on adversarial samples. Our research incorporates a realistic threat model that takes into account the knowledge and capabilities of potential attackers, following the taxonomy outlined in Machado et al. (2021). We conduct a rigorous evaluation of the proposed defense using a dataset collected from a real-world power system testbed, offering an authentic representation of the operational environment. The key contributions of this paper are as follows:

1. Introduction of the *RADIANT* IDS, a novel reactive adversarial defense mechanism designed to effectively identify disguised adversarial attacks without the need for prior training on adversarial samples.
2. Generation of adversarial samples using two state-of-the-art attack methods: the hopskipjump attack (Chen et al., 2020) and the zoo attack (Chen et al., 2017), enabling comprehensive evaluation of the defense mechanism.
3. Performance evaluation of the *RADIANT* IDS against various baseline classifiers, using a real-world power system testbed dataset and employing four distinct evaluation metrics to assess its effectiveness.
4. Investigation of the adversarial robustness of the *RADIANT* IDS under two different adversarial attack scenarios, within the context of a realistic threat model.
5. Comparative assessment of the adversarial robustness of the *RADIANT* IDS against leading state-of-the-art adversarially robust IDSs, including those proposed by Anthi et al. (2021), Roshan et al. (2023), Debicha et al. (2021) and Pawlicki et al. (2020), demonstrating the superior versatility and effectiveness of the proposed reactive defense mechanism.

The research study outlined in this paper follows a structured methodology, as shown in Fig. 1. The methodology consists of the following steps:

1. **Dataset Preprocessing and Partitioning**: The power system dataset was first preprocessed and then divided into training and testing subsets with a 2:1 ratio.
2. **Model Training**: Three classifiers were trained on the training set: a simple Autoencoder-based model (AE), the proposed *RADIANT* IDS, and a baseline Random Forest classifier.
3. **Performance Evaluation**: The performance of these classifiers was evaluated on the test set using four standard evaluation metrics: Precision, Recall, F1 Score, and Accuracy.
4. **Adversarial Sample Generation**: Adversarial samples were generated using two distinct attack techniques: the hopskipjump attack and the zoo attack.
5. **Adversarial Robustness Assessment**: The trained classifiers were re-evaluated on these adversarial samples to assess their robustness against adversarial attacks.
6. **Comparison with State-of-the-Art Defenses**: To facilitate comparison with existing adversarial defense approaches, including those proposed by Anthi et al. (2021), Roshan et al. (2023), Debicha et al. (2021) and Pawlicki et al. (2020), adversarial samples were generated using both hopskipjump and zoo attacks on the training set.

7. **Training State-of-the-Art Classifiers**: The state-of-the-art classifiers were trained using the original training set along with the generated adversarial samples.
8. **Evaluation of State-of-the-Art Classifiers**: The performance of these classifiers was evaluated on both the test dataset and adversarial samples generated from previously unseen test instances. This enabled a comparative analysis of their adversarial robustness relative to the proposed *RADIANT* IDS.

The subsequent sections of this paper are organized as follows: In Section 2, we review the existing research on ML-based IDSs in the context of ICS, including their adversarial defenses. Section 3 provides an overview of the ICS case study and the dataset used in this study. Section 4 outlines the adversarial threat model, which is crucial for assessing IDS vulnerability and the effectiveness of defense mechanisms. In Section 5, we present the details of the proposed *RADIANT* IDS, including its architecture and training process. Section 6 describes the evaluation framework, which is key to assessing performance and enabling comparative analysis. Section 7 presents the results and conducts a detailed comparative analysis. Finally, the concluding Sections 8 and 9, summarize our findings, discuss potential limitations, and propose directions for future research.

2. Related work

Numerous studies have demonstrated that machine learning techniques can significantly enhance the performance of IDSs. However, the success of ML-based IDSs largely depends on the quality and diversity of their training data (Picard et al., 2020). These systems often face challenges when encountering data patterns or regions not adequately represented during training. Such "blind spots" in the data can lead to inaccurate predictions, particularly when new or previously unseen inputs fall within these regions.

Adversarial attacks exploit these vulnerabilities by introducing subtle perturbations to input data, deceiving ML models into making incorrect predictions. This tactic allows attackers to bypass detection or cause misclassifications. Therefore, evaluating IDS performance solely on traditional test datasets is insufficient; it is crucial to assess their resilience to adversarial attacks to ensure robust and reliable operation.

Recent research has investigated the impact of adversarial attacks on IDSs (Trivedi et al., 2023; Anthi et al., 2021; Wali and Khan, 2021). For instance, Trivedi et al. (2023) analyzed the effects of adversarial samples generated through the Fast Gradient Sign Method (FGSM), exposing the challenges in maintaining accuracy under attack. Similarly, Alarab and Prakoonwit (2023) demonstrated adversarial vulnerabilities using an ICS dataset, revealing how a well-trained LSTM classifier struggled to distinguish anomalies in adversarial settings. These findings underscore the urgency for comprehensive vulnerability assessments and the development of more effective defense mechanisms.

In response, Anthi et al. (2021) proposed a proactive approach using adversarial retraining to enhance system robustness. Building on this concept, Roshan et al. (2023) introduced a defense mechanism for deep learning-based Network Intrusion Detection Systems (NIDS) that mitigates attacks such as FGSM, Jacobian Saliency Map Attack (JSMA), Projected Gradient Descent (PGD), and Carlini & Wagner (C&W). While adversarial retraining improved robustness against known threats, it faced significant limitations in generalizing to unseen or more sophisticated attacks, such as C&W. This highlights the persistent vulnerability of adversarially retrained systems to novel attack strategies, emphasizing the critical need for more adaptive and versatile solutions.

Debicha et al. (2021) explored the transferability of adversarial attacks and developed an ensemble-based IDS combining five ML models (SVM, Decision Tree, Logistic Regression, Random Forest, and LDA) through majority voting. They introduced a "Detect & Reject" mechanism by adding a third class ("adversarial") during training to identify and reject adversarial inputs. While this approach improved robustness, particularly for non-differentiable models, its reliance on accurate adversarial data generation and ensemble complexity limited its overall effectiveness. As Machado et al. (2021) noted, generating all possible adversarial data types is impractical, leaving systems vulnerable to unforeseen attacks and highlighting the need for more reactive defense strategies.

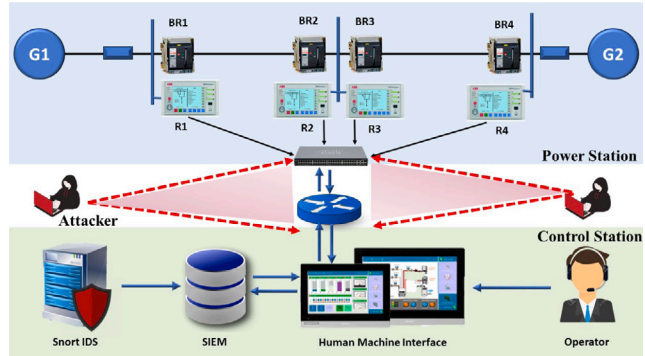

Fig. 2. The power system testbed, collaboratively developed by Mississippi State University and Oak Ridge National Laboratory, features a scaled-down power system framework, providing a representative model for larger power systems. This testbed serves as the foundation for generating the datasets essential for the experiments in this study (MSU and ORNL, 2014).

Pawlicki et al. (2020) proposed a secondary adversarial detector trained to identify adversarial samples by analyzing neuron activations in the primary IDS model for benign, standard attack, and adversarial samples. Using adversarial examples generated by FGSM, Basic Iterative Method (BIM), C&W, and PGD, they trained the secondary classifier to distinguish these patterns. While this method achieved high recall for known attack types, its effectiveness decreased against novel or unexpected adversarial techniques. Frequent retraining with newly generated adversarial samples was required, making the process resource-intensive and reducing adaptability.

Although adversarial retraining and detection mechanisms have shown promise in enhancing IDS robustness, they present notable challenges. Adversarial training alone fails to guarantee robustness against diverse attack algorithms, as it demands an extensive and varied training dataset comprising numerous adversarial samples across different methods and perturbation levels (Machado et al., 2021). This dependency increases training time and computational overhead. To address these limitations, our proposed scheme focuses on developing a more efficient defense mechanism that minimizes reliance on adversarial training while enhancing resilience and computational efficiency.

3. Case study: Industrial control system in the power sector

The power system is a vital component of critical infrastructure, making it a relevant case study for exploring the challenges and vulnerabilities of ICS when subjected to AML attacks. ICS technologies often play a crucial role in controlling and monitoring power systems, making them susceptible to emerging AML threats. Mississippi State University and Oak Ridge National Laboratory implemented a scaled-down version of a power system framework, serving as a representative example of larger power systems (Pan et al., 2015). While smaller in scale, this framework captures the core functions of power systems. Fig. 2 offers a detailed illustration of the power system framework configuration and the components used to generate datasets supporting the experiments in this paper. These components, within the power system context, offer valuable insights into the broader domain of ICS and its susceptibility

Table 1
A list of attack scenarios covered in the power system dataset.

Data label	Category	Scenario	Brief description	Data instances
Benign	No event	Normal operation Maintenance activities Faults	Electrical load variations Line corrective or Preventive maintenance Natural Ground Faults	22,714
Malicious	Attack event	Data injection Command injection Settings manipulation	Ground Faults with tripping command Remote tripping command sent to one relay Remote tripping command sent to two relays Single relay was disabled and fault Two relays were disabled and fault Two relays were disabled and Line maintenance	55,663

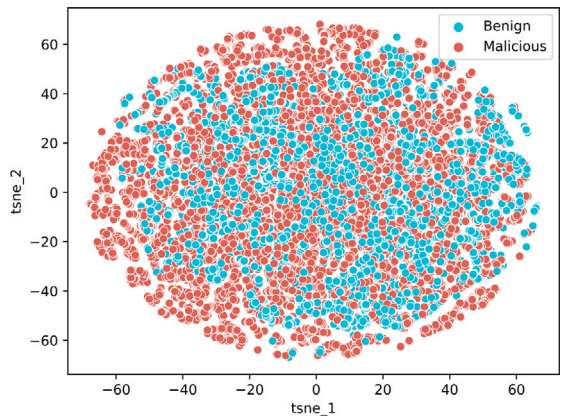

Fig. 3. t-SNE visualization of benign and malicious data.

Table 2
Description of power system dataset features.

Features type	Features description
PMU measurements	3-Phase voltage magnitude 3-Phase voltage angle 3-Phase current magnitude 3-Phase current angle Pos.-Neg.-Zero Seq. 3-Phase voltage magnitude Pos.-Neg.-Zero Seq. 3-Phase voltage angle Pos.-Neg.-Zero Seq. 3-Phase current magnitude Pos.-Neg.-Zero Seq. 3-Phase current angle Frequency Frequency delta Impedance Impedance angle Status
PMU logs	Snort log Control panel log Relay logs

to AML attacks, aiding in the development of more resilient defense mechanisms.

Specifically, the power system components depicted in Fig. 2 encompass:

- G1 and G2, serving as the primary generators.
- Relays (R1 to R4), representing Intelligent Electronic Devices (IEDs) responsible for controlling the operation of breakers (BR1 to BR4). These breakers are automated electrical switches designed to safeguard electrical circuits from damage due to excessive current resulting from overload or short circuit conditions.
- Each IED is dedicated to a specific breaker (for instance, R1 manages BR1, R2 handles BR2, and so forth).
- The IEDs employ a distance protection scheme, which triggers the breaker in response to detected faults, whether they are valid or invalid, as they lack internal validation to distinguish between the two.
- Operators can manually issue commands to the IEDs for tripping the breakers. This override is particularly useful during maintenance on power lines or other system components.
- The testbed incorporates additional network monitoring devices such as SNORT and Security Information and Event Management (SIEM) system.

3.1. Dataset description

The dataset used in this research, containing both benign and malicious data points, was derived from the power system testbed by MSU and ORNL (2014). These data points were subsequently classified into three main categories: “no event” instances, “natural event” instances, and “attack event” instances. The “no event” and “natural event” instances are grouped together to represent benign activities. The malicious data was generated by simulating attacks from five distinct scenarios on the power system. Descriptions of these scenarios are provided in Table 1.

3.2. Features description

A phasor measurement unit (PMU), also known as a synchrophasor, is a device employed to assess the electrical wave characteristics within an electricity grid. It synchronizes measurements using a shared time source. The dataset, primarily derived from the testbed, predominantly includes parameters captured by PMUs (MSU and ORNL, 2014). Among the dataset’s total of 128 features, 116 stem from PMUs, with each PMU contributing 29 distinct features, as detailed in Table 2. Furthermore, the dataset includes 12 additional features, such as control panel logs, Snort alerts, and relay logs, each associated with an individual PMU.

3.3. Dataset visualization

t-Distributed Stochastic Neighbor Embedding (t-SNE) is a widely used dimensionality reduction technique in machine learning and data visualization (Chan et al., 2019). It aims to capture complex relationships between data points in high-dimensional spaces and represent them in lower dimensions, typically 2D or 3D, while preserving their pairwise similarities. In the context of the power system dataset, t-SNE was applied to provide a 2D visualization of the data. In Fig. 3, the resulting plot illustrates that the benign and malicious data instances within the dataset do not exhibit linear separability. This means that a simple linear boundary cannot effectively distinguish between the two classes, indicating that more sophisticated non-linear classifiers or anomaly detection methods may be necessary to accurately discern between benign and malicious instances in this power system dataset. Such inseparability carries significant implications, as it implies that malicious data could potentially be disguised as benign, posing a security risk. Therefore, classifiers designed for power system security must exhibit robustness against such adversarial attempts to ensure accurate threat detection and system protection.

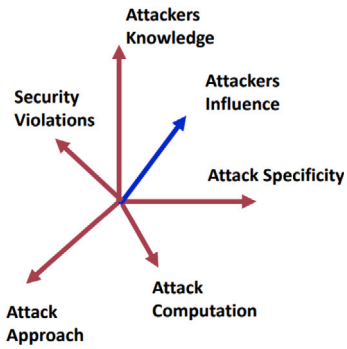

Fig. 4. Adversarial attack and attacker taxonomy.

4. Threat model

A threat model typically outlines key elements, such as the attacker's knowledge of the classifier, their objectives in launching the attack, and the methods they use during the adversarial attack process. Understanding the attacker's capabilities and the strategies employed in generating adversarial attacks is essential for designing a defense system that can provide security guarantees under predefined conditions (Carlini et al., 2019). For this research, we consider a threat model based on the concepts and definitions from previous works (Akhtar and Mian, 2018; Brendel et al., 2017; Barreno et al., 2010; Kumar and Mehta, 2017; Xiao, 2017; Yuan et al., 2019). Consequently, the threat model examined in this study categorizes the adversarial landscape along six distinct axes, as illustrated in Fig. 4, which are further discussed in the following sections.

4.1. Attacker's influence

This dimension clarifies the extent of influence the attacker has in manipulating the model. There are two categories under this dimension: poisoning, where the attacker can manipulate the data during training, and evasion, where the attacker targets the model during inference. Our threat model assumes that the attacker can only influence the model during inference, performing evasive attacks.

4.2. Attacker's knowledge

Our threat model is based on the grey-box attack scenario, initially proposed by Meng and Chen (2017). In grey-box attacks, the attacker has access to the classification model but lacks information about the defense method. Grey-box attacks provide an intermediate level of threat, more severe than black-box attacks but not as advantageous for the attacker as white-box scenarios where they have full knowledge of the defense method.

4.3. Security violation

Security violations are associated with the attacker's objectives when conducting an adversarial attack. These violations can affect the integrity, availability, and privacy of the targeted classifiers. In our defined threat model, the adversary's goal is to compromise integrity by disguising malicious data so that it is classified as benign, thus violating the system's integrity.

4.4. Attack specificity

This dimension determines whether the attack targets specific classes. In our threat model, the adversary launches targeted attacks by manipulating only malicious data to make it appear benign.

4.5. Attack computation

This dimension relates to the algorithms used to compute perturbations. They can be sequential or iterative. Sequential algorithms compute the perturbation in a single step, while iterative algorithms require multiple iterations to craft the perturbation. Although iterative algorithms have a higher computational cost, they often generate smaller and more effective perturbations for fooling classification models. Our threat model assumes that the adversary possesses sufficient computational power to craft sophisticated adversarial samples using iterative methods.

4.6. Attack approach

Various approaches can be employed to develop adversarial samples. In our threat model, we employ two distinct approaches to generate these samples, both of which are described in the following subsection.

4.6.1. Zeroth-Order Optimization (ZOO) attack

Zeroth-Order Optimization (ZOO) adversarial attacks use mathematical optimization techniques to craft adversarial examples without direct access to gradients (Chen et al., 2017). Some key equations and concepts commonly involved in ZOO attacks are presented below:

Perturbation: In ZOO attacks, the goal is to find a perturbation, often denoted as δ, to add to the original input sample x to create the adversarial example x_{adv}:

$$x_{\text{adv}} = x + \delta \tag{1}$$

Objective Function: The objective function $J(\delta)$ tries to minimize the model's confidence in the correct class and maximize its confidence in the target class. This is done by iteratively adding perturbations (δ) to the input image until the model's predicted output for the target class is maximized. For targeted attacks, this may be written as:

$$J(\delta) = -f(x+\delta)_{\text{target}} \tag{2}$$

Here, $f(x)$ is the model's prediction for input x, and $_{\text{target}}$ denotes the target class.

Constraint Function: ZOO attacks often operate under constraints to ensure the perturbation δ remains within certain bounds. For example, a common constraint is to limit the L_2 norm of δ:

$$\|\delta\|_2 \leq \epsilon \tag{3}$$

where ϵ is a user-defined hyperparameter controlling the perturbation magnitude.

Optimization Objective: The primary objective in ZOO attacks is to find the perturbation δ that minimizes the objective function $J(\cdot)$ while satisfying the constraints. This is typically formulated as an optimization problem:

$$\begin{aligned} &\text{minimize } J(\delta) \\ &\text{subject to } \|\delta\|_2 \leq \epsilon \end{aligned} \tag{4}$$

Optimization Techniques: ZOO attacks use various optimization methods to solve the above problem. These methods can include evolutionary algorithms, Bayesian optimization, coordinate descent, or other black-box optimization techniques. The choice of optimization method can influence the specific equations used.

Gradient Estimation: Since ZOO attacks lack access to gradients, they use surrogate methods to estimate gradient information. One common approach is to use finite differences to approximate the gradient of the objective function:

$$\nabla J(\delta) \approx \frac{J(\delta+\Delta) - J(\delta-\Delta)}{2\Delta} \tag{5}$$

where Δ is a small perturbation.

Step Size: During optimization, a step size or learning rate (α) is applied to the gradient estimate to update the perturbation:

$$\delta_{\text{new}} = \delta - \alpha \nabla J(\delta) \tag{6}$$

Iterations: ZOO attacks perform multiple iterations of the optimization process, gradually refining the perturbation until a suitable adversarial attack is generated. Such sophisticated adversarial samples can easily bypass ML based IDSs.

4.6.2. HopSkipJump attack

The HopSkipJump Attack requires significantly fewer model queries than several state-of-the-art decision-based adversarial attacks (Chen et al., 2020). It also achieves competitive performance in attacking several widely-used defense mechanisms. The HopSkipJump attack works by iteratively searching for a perturbed input that is closer to the decision boundary of the target model and that is also misclassified by the model. The attack starts with a random input and then iteratively applies two types of perturbations to the input:

The first type of perturbation is known as Hop perturbation. The hop perturbation is a simple addition of a random noise vector to the input. The noise vector is sampled from a Gaussian distribution with zero mean and a standard deviation that is inversely proportional to the distance between the input and the decision boundary of the target model. Consequently, these perturbations move the input closer to the decision boundary of the target model.

$$\text{Hop}(x) = x + \sigma \cdot \text{noise} \tag{7}$$

Here,x is input vector,noise is a random noise vector sampled from a Gaussian distribution with zero mean and σ is the standard deviation.

The second type of perturbation is known as Skip perturbation. These perturbations are based on a more complex operation that involves first finding the closest class to the input on the decision boundary of the target model. The input is then moved to the closest class on the decision boundary of the target model that is different from the current class of the input.

$$\text{Skip}(x) = \text{argmin}_{y \neq \text{class}(x)} \|x - \text{decision boundary}(y)\| \tag{8}$$

Here, class(x) is the predicted class of input x by the target model and decision boundary(y) is the decision boundary of the target model for class y. This attack terminates when either a misclassified input is found or a predetermined number of iterations is reached.

5. Methodology

To enhance the resilience of ML-based IDS against adversarial threats, we propose a novel reactive defense mechanism through the RADIANT module. This module, positioned between the preprocessing phase and the machine learning classifier, significantly improves adversarial robustness. As illustrated in Fig. 5, the RADIANT module can be integrated with any baseline classifier; in our study, it is combined with a Random Forest classifier. The resulting RADIANT IDS, as depicted in Fig. 5, consists of several critical components, which are outlined in the subsequent headings.

5.1. Dataset preprocessing

The foundation of our RADIANT IDS is laid upon the ICS dataset described in Section 3. During preprocessing, data from all fifteen CSV files is consolidated and scrutinized for erroneous and corrupt instances. Labels in the dataset are encoded into integers, designating '0' for benign class and '1' for malicious or attack instances.

5.2. Data transformation

After the initial preprocessing steps, the dataset is subjected to scaling through the application of the MinMaxScaler from the sklearn library. This transformation standardizes feature values to a specified range, typically between 0 and 1. It ensures that features with varying units or scales do not dominate the learning process, rendering the data suitable for diverse machine learning algorithms.

5.3. Data splitting

Subsequent to the preprocessing and transformation stages, the entire dataset was partitioned into training and testing subsets. A 33% portion of the data was allocated for testing to evaluate the performance of the trained classifier, while the remaining 67% was designated for training and validation purposes. The class distribution in this train-test split is illustrated in Fig. 6.

5.4. RADIANT module

The RADIANT module introduces a novel, reactive defense mechanism aimed at enhancing the adversarial robustness of ML-based IDS. This section outlines the key components of the RADIANT module, its feature extraction process, and the rationale behind its defense capabilities against adversarial attacks, with an explanation based on the boundary tilting hypothesis (Machado et al., 2021).

5.4.1. Key components of the RADIANT module

The RADIANT module relies on three parallel autoencoders, each trained on different data types, enabling the system to respond dynamically to various adversarial and attack scenarios. These include:

- **Denoising Autoencoder (DAE)**: Trained on both benign and malicious data, the DAE's primary function is to remove noise or perturbations, enhancing the system's ability to recover clean features from adversarial inputs.
- **Benign-only Autoencoder**: Trained exclusively on benign data, this AE captures the distribution of normal behavior and provides a baseline for distinguishing between benign and malicious activities.
- **Attack-only Autoencoder**: This AE is trained solely on attack data, enabling the system to recognize features indicative of cyber attack patterns and detect attack-induced anomalies.

All three autoencoders share an identical architecture, as outlined in Table 3. Each autoencoder comprises several fully connected layers, batch normalization layers, and dropout layers for regularization. The encoder compresses the input into a 16-dimensional latent space, which is subsequently reconstructed by the decoder. This architecture allows the autoencoders to learn compact, informative representations of the data, which are essential for detecting adversarial perturbations.

5.4.2. Feature extraction: Latent space and reconstruction error

Each autoencoder extracts latent space representations, denoted as z_{Benign}, z_{DAE}, and z_{Attack}, which capture the essential features of the input data in a compact, low-dimensional space. These representations are instrumental in identifying anomalies by filtering out noise and focusing on the core characteristics of the data.

In addition to latent space features, the reconstruction error plays a critical role in feature extraction. For each autoencoder, the reconstruction error $\mathcal{L}_{\text{Reconstruction}}$ quantifies how well the model can reconstruct the input data and measures the deviation of the input from the learned distribution for each data type (benign, adversarial, or attack). The reconstruction error is defined as:

$$\mathcal{L}_{\text{Reconstruction}} = \|x - f_{\text{AE}}(x)\|^2 \tag{9}$$

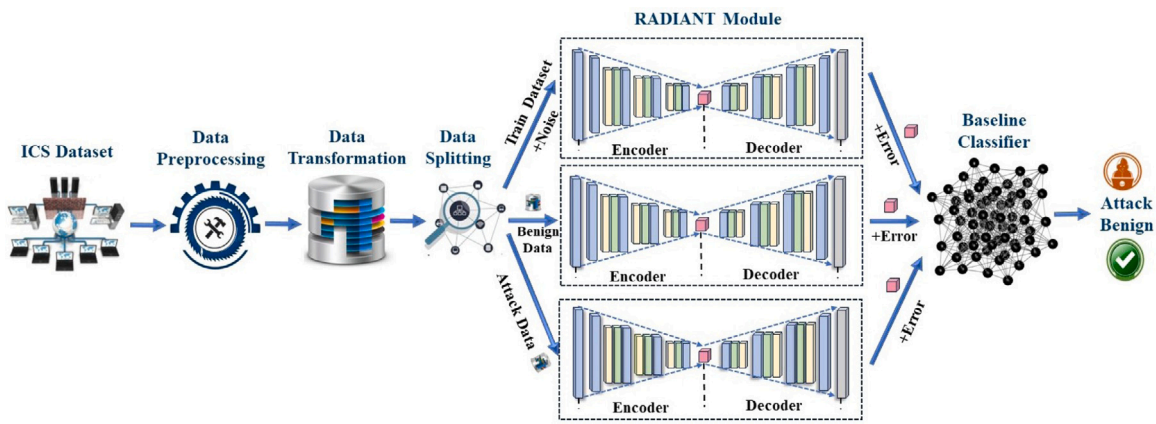

Fig. 5. An overview of the RADIANT IDS methodology - A novel approach to enhance adversarial robustness in machine learning-based intrusion detection systems. This figure illustrates the central role of the RADIANT module, strategically positioned between the data preprocessing stage and the machine learning classifier (Random Forest), reinforcing the IDS against adversarial attacks. The comprehensive RADIANT IDS consists of several key components, including dataset preprocessing, transformation, and data splitting. Additionally, it incorporates an innovative trio of autoencoders within the RADIANT module for robust feature extraction, along with the baseline Random Forest classifier, which differentiates between normal and malicious network traffic.

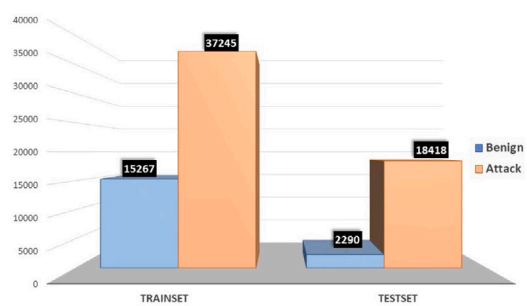

Fig. 6. Benign and attack data instances within both the training and testing datasets.

Table 3
Autoencoder model architecture.

Layer type	Layer description	Output shape
Encoder	Input layer	(None, 128)
	Batch normalization	(None, 128)
	Dense	(None, 102)
	Dense	(None, 84)
	Dropout (0.1)	(None, 84)
	Batch normalization	(None, 84)
	Dense	(None, 64)
	Dropout (0.1)	(None, 64)
	Batch normalization	(None, 64)
	Dense	(None, 44)
	Dropout (0.1)	(None, 44)
	Batch normalization	(None, 44)
Latent space	Dense	(None, 16)
Decoder	Dense	(None, 44)
	Dropout (0.1)	(None, 44)
	Batch normalization	(None, 44)
	Dense	(None, 64)
	Dropout (0.1)	(None, 64)
	Batch normalization	(None, 64)
	Dense	(None, 84)
	Dropout (0.1)	(None, 84)
	Batch normalization	(None, 84)
	Dense	(None, 102)
	Dense	(None, 128)

where $f_{\mathrm{AE}}(x)$ represents the output of the respective autoencoder (DAE, Benign-only AE, or Attack-only AE), and x is the input data. This error captures the extent to which the input deviates from the expected feature distribution for each class of data.

Training curves for all autoencoders, shown in Fig. 7, demonstrate the convergence of the reconstruction error during training. These curves confirm that the models generalize effectively across both benign and malicious data, indicating their ability to adapt and learn robust features from diverse inputs.

5.4.3. Combined feature representation

The feature space for the RADIANT module is created by combining the latent space representations and reconstruction errors from all three autoencoders. The resulting feature vector, denoted as z_{RADIANT}, integrates both high-level data representations and error metrics, enhancing its resilience to adversarial perturbations:

$$z_{\mathrm{RADIANT}} = \left[z_{\mathrm{Benign}}, z_{\mathrm{DAE}}, z_{\mathrm{Attack}}, \mathrm{Error}_{\mathrm{Benign}}, \mathrm{Error}_{\mathrm{DAE}}, \mathrm{Error}_{\mathrm{Attack}}\right] \tag{10}$$

This comprehensive feature vector, z_{RADIANT}, incorporates multiple levels of information about the input data, thereby making it more robust to adversarial inputs, attack data, and natural deviations. Then this enriched feature set can be fed into any baseline classifier, such as Random Forest, to improve the classifier's ability to distinguish between benign and malicious data.

5.4.4. Defensive role: Mitigating adversarial perturbations

The primary objective of the RADIANT module is to mitigate the impact of adversarial perturbations, ensuring that the input data remains on the correct side of the classifier's decision boundary B. In traditional adversarial machine learning settings, perturbations ϵ are added to inputs in order to cause misclassification. However, by processing the input through the RADIANT module, the perturbation is mitigated in the feature space before it influences the decision-making process.

Mathematically, let $\hat{x} = x + \epsilon$ denote an adversarially perturbed input, where x is the original input and ϵ represents the adversarial perturbation. The goal of the RADIANT module is to map the perturbed input $\hat{x}$ to a more robust feature space z_{RADIANT}, where the perturbation ϵ has a diminished effect on the final decision:

$$z_{\mathrm{RADIANT}} = f_{\mathrm{RADIANT}}(x + \epsilon) \approx f_{\mathrm{RADIANT}}(x) \tag{11}$$

By passing the perturbed input through the RADIANT module, the resulting feature set z_{RADIANT} becomes less influenced by the perturbation, ensuring that the input remains on the correct side of the decision boundary B. This is achieved by using the autoencoders to map the input to a space where adversarial perturbations are less likely to shift the decision boundary. The denoising role of the DAE helps to ensure that the features extracted from both benign and adversarial data are

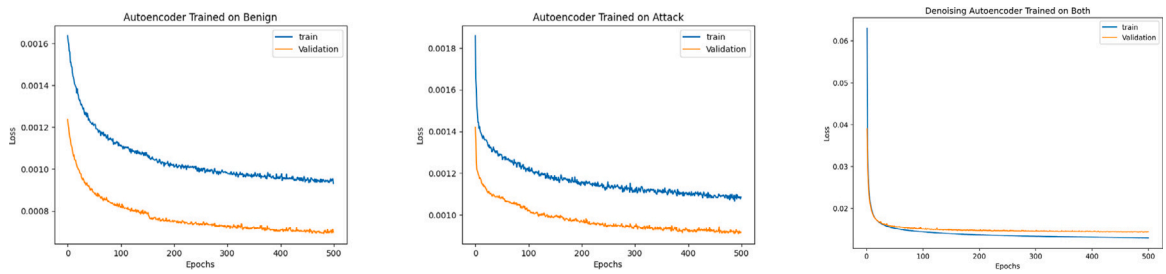

Fig. 7. Training and validation loss curves of RADIANT autoencoders.

clean, which in turn reinforces the system's resilience to adversarial manipulations.

This process effectively reduces the impact of adversarial perturbations, helping the classifier maintain accurate classification despite the presence of noise and attack-induced modifications. Consequently, the RADIANT module enhances the robustness of machine learning-based intrusion detection systems, ensuring that adversarial attacks have minimal effect on the classifier's decision-making process.

5.5. Baseline classifier

After processing the input through the RADIANT module, the resulting feature vector z_{RADIANT} is passed to a baseline classifier. The RADIANT module is designed to integrate effectively with a variety of baseline classifiers, including neural networks and deep learning architectures, with its primary objective being to enhance the adversarial robustness of these classifiers without requiring adversarial training. For this study, we employed the Random Forest Classifier (RFC) as the baseline classifier, as it is a widely used machine learning algorithm for both classification and regression tasks across various domains (Wali and Khan, 2022; Farrukh et al., 2021). RFC works by constructing an ensemble of decision trees, each trained on a distinct subset of the data, introducing randomness during the tree-building process. The final prediction is determined by aggregating the predictions from individual trees, typically using voting for classification tasks or averaging for regression tasks. To ensure a fair and consistent comparison with previous methods, such as those presented in Anthi et al. (2021), we used the Random Forest Classifier from the sklearn library with default parameters.

6. Evaluation framework

To thoroughly evaluate the effectiveness and adversarial resilience of the RADIANT module, a comprehensive evaluation framework is proposed. This framework is based on established guidelines (Carlini et al., 2019) and emphasizes conducting rigorous sanity checks to assess the module's performance. The framework is organized into the following key sections:

6.1. Performance evaluation on test set

Evaluating the performance of the RADIANT module on a standard test set is crucial to achieving an optimal balance between adversarial robustness and accuracy on legitimate data. While enhancing adversarial resilience is a critical security requirement, an overemphasis on this aspect can result in significant accuracy degradation on legitimate data, which is impractical in contexts where adversarial attacks are infrequent. For reactive defenses like RADIANT, it is imperative to thoroughly analyze the impact of the proposed defense mechanism on the overall performance of the IDS under non-adversarial conditions, ensuring that the implementation of adversarial defenses does not undermine the system's effectiveness on legitimate traffic. Therefore, we will evaluate the performance of all IDS on the test set to ensure a comprehensive assessment in our evaluation.

6.2. Ablation study

To systematically evaluate the effectiveness of the proposed reactive defense mechanism, an ablation study will be conducted to isolate and measure its impact. This study involves removing the protective mechanism from the IDS and assessing the baseline classifier's performance on adversarial samples, following the guidelines of Carlini et al. (2019). The goal is to determine whether the baseline IDS exhibits any inherent robustness against adversarial threats. Additionally, we will compare RADIANT IDS with an alternative system referred to as the Autoencoder-based (AE-based) classifier, which adopts a simplified architecture by employing a single autoencoder during processing. By analyzing these comparisons, we aim to highlight the specific contributions of the RADIANT IDS's reactive defense mechanism in enhancing adversarial resilience, thereby validating its design and practical value.

6.3. Comparison with state-of-the-art approaches

To benchmark the RADIANT IDS against current advancements in adversarial defense mechanisms, we will conduct a detailed comparative analysis with state-of-the-art methods. This includes proactive defenses such as the adversarially retrained Random Forest model proposed by Anthi et al. (2021) and deep neural architectures by Roshan et al. (2023). Additionally, we will evaluate RADIANT against ensemble-based defenses, such as the majority-voting approach introduced by Debicha et al. (2021), as well as their detect-and-reject mechanism, which adds an "adversarial" label class to the IDS. Another comparison involves the secondary adversarial detector by Pawlicki et al. (2020), which trains classifiers using neuron activation data to distinguish between legitimate and adversarial samples. Through this comparative evaluation, we aim to demonstrate the efficacy and competitiveness of the RADIANT IDS in addressing adversarial threats while maintaining high performance.

6.4. Evaluation metrics

Our evaluation framework employs four different evaluation metrics to comprehensively assess the performance of our proposed approach. These metrics provide insights into various aspects of the model's effectiveness in detecting intrusions and its resilience against adversarial attacks. We use the following evaluation measures:

- **Precision (P)**: Precision measures the accuracy of positive predictions, representing the ratio of true positive predictions to the total positive predictions.

$$P = \frac{TP}{TP + FP} \tag{12}$$

  Here TP is True Positive and FP is False Positive
- **Recall (R)**: Recall quantifies the model's ability to identify all relevant instances, indicating the ratio of true positive predictions to the total actual positives.

$$R = \frac{TP}{TP + FN} \tag{13}$$

  Here TP is True Positive and FN is False Negative

Table 4
Performance evaluation on the test set.

ML-based IDS	Precision	Recall	F1 Score	FPR
Baseline IDS	0.949	0.938	0.944	0.015
AE-based IDS	0.946	0.936	0.941	0.018
Anthi et al. (2021)	0.947	0.934	0.941	0.017
Roshan et al. (2023)	0.928	0.938	0.933	0.038
Ensemble (Debicha et al., 2021)	0.960	0.952	0.956	0.013
Detect & Reject (Debicha et al., 2021)	0.930	0.949	0.939	0.037
Pawlicki et al. (2020)	0.890	0.947	0.918	0.092
Proposed IDS	0.947	0.937	0.942	0.016

- **F1 Score**: The F1 Score provides a balanced measure between Precision and Recall, calculated as the harmonic mean of Precision and Recall.

$$F1 = \frac{2 \cdot P \cdot R}{P + R} \tag{14}$$

- **False Positive Rate (FPR)**: The False Positive Rate in an IDS represents the likelihood of misclassifying a benign (normal) instance as an attack, leading to false alarms. In ICS, a high FPR can cause unnecessary alerts, disrupting critical processes, increasing operational costs, and straining response teams. Therefore, maintaining a low FPR is crucial to ensuring the IDS remains both effective and practical, minimizing unwarranted interventions while preserving the reliability of normal system operations.

$$FPR = \frac{FP}{FP + TN} \tag{15}$$

Here FP is False Positive, and TN is True Negative

Through these metrics, we aim to gain a comprehensive understanding of how our RADIANT IDS performs under different conditions and comparative analyses with baseline classifiers.

7. Results

This section evaluates the performance of the RADIANT IDS by comparing it with baseline classifiers and state-of-the-art proactive defense mechanisms across three scenarios. First, the IDS is assessed on legitimate, unperturbed data samples from the test set to establish baseline performance under normal conditions. Subsequently, its resilience is tested against two adversarial attacks: HopSkipJump and Zoo. For each attack, the evaluation begins with a comparison to baseline classifiers, including undefended or partially defended models as part of an ablation study, to isolate the impact of the proposed defense mechanism. Finally, the IDS is benchmarked against state-of-the-art approaches to demonstrate its effectiveness against advanced adversarial threats. The results for each scenario are detailed below.

7.1. Performance evaluation on test set

To ensure that the proposed IDS maintains high accuracy on normal, unperturbed traffic, we evaluate its performance on the test set, which consists of benign and attack samples without adversarial perturbations. This evaluation confirms that the adversarial defense mechanisms do not compromise baseline accuracy.

Table 4 presents the performance of the proposed IDS alongside baseline and state-of-the-art classifiers. The results demonstrate that most IDSs achieve consistently high performance across key evaluation metrics, including precision, recall, F1-score, and FPR. The proposed IDS performs competitively, aligning with state-of-the-art approaches, while ensemble-based methods exhibit slightly superior performance, reinforcing their ability to handle complex attack scenarios. These findings highlight not only the robustness of the proposed IDS but also its ability to operate effectively under normal conditions, ensuring that the integration of adversarial defenses does not degrade standard detection performance.

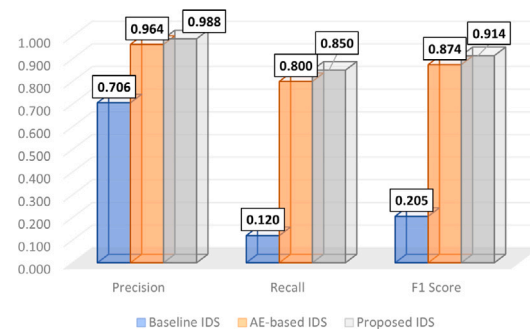

Fig. 8. Ablation study results under the HopSkipJump attack, comparing the baseline classifier (undefended), AE-based classifier (partially defended), and the proposed RADIANT IDS.

7.2. Adversarial robustness evaluation on Hopskipjump attack

In alignment with our threat model, we evaluated the RADIANT IDS under the HopSkipJump adversarial attack, where adversarial samples were carefully crafted to mislead the system into classifying them as benign. This evaluation was carried out in two phases: first, an ablation study to assess the performance of undefended and partially defended models, followed by a comparison with state-of-the-art defense mechanisms.

The ablation study, illustrated in Fig. 8, revealed significant insights into the impact of defense mechanisms. The baseline classifier, without any defense, exhibited poor performance, with an F1 score of only 20.5%, underscoring its vulnerability to adversarial manipulation. In contrast, the AE-based classifier, which incorporates a partial defense, demonstrated a notable improvement, achieving an F1 score of 87.4%. This highlights the effectiveness of even basic defense strategies in enhancing the model's resilience to adversarial attacks.

Building on this, we next compared the performance of RADIANT IDS with state-of-the-art defense approaches, as summarized in Table 5. The proposed RADIANT IDS outperformed all other models, achieving the highest F1 score of 91.4% and FPR of only 1.2%, demonstrating its superior robustness against the HopSkipJump attack. Among the state-of-the-art models, the detect-and-reject mechanism (Debicha et al., 2021) achieved an F1 score of 86.5%, and the ensemble-based approach reported an F1 score of 76.4%. On the other hand, adversarially retrained classifiers (Anthi et al., 2021) and secondary adversarial detectors (Pawlicki et al., 2020) showed relatively lower performance, further emphasizing the efficacy of the RADIANT IDS in defending against this specific attack. These results confirm that RADIANT IDS is the most robust defense mechanism in this evaluation.

7.3. Adversarial robustness evaluation on Zoo attack

The robustness of the RADIANT IDS was further evaluated under the ZOO adversarial attack. As shown in Fig. 9, the ablation study highlights the vulnerabilities of the baseline classifier, which achieved an F1 score of only 17.1%, indicating its susceptibility to adversarial manipulations. In contrast, the AE-based classifier, with its partial

Table 5
Performance comparison on HopSkipJump attack samples.

ML-based IDS	Precision	Recall	F1 Score	FPR
Anthi et al. (2021)	0.636	0.350	0.452	0.205
Roshan et al. (2023)	0.833	0.750	0.789	0.153
Ensemble (Debicha et al., 2021)	0.872	0.680	0.764	0.102
Detect & Reject (Debicha et al., 2021)	0.952	0.792	0.865	0.043
Pawlicki et al. (2020)	0.933	0.700	0.800	0.054
Proposed IDS	**0.988**	**0.850**	**0.914**	**0.012**

Table 6
Performance comparison on ZOO attack samples.

ML-based IDS	Precision	Recall	F1 Score	FPR
Anthi et al. (2021)	0.935	0.430	0.589	0.039
Roshan et al. (2023)	0.800	0.600	0.686	0.158
Ensemble (Debicha et al., 2021)	0.917	0.550	0.688	0.056
Detect & Reject (Debicha et al., 2021)	0.940	**0.780**	0.852	0.052
Pawlicki et al. (2020)	**0.984**	0.620	0.761	0.019
Proposed IDS	0.976	0.766	**0.859**	**0.014**

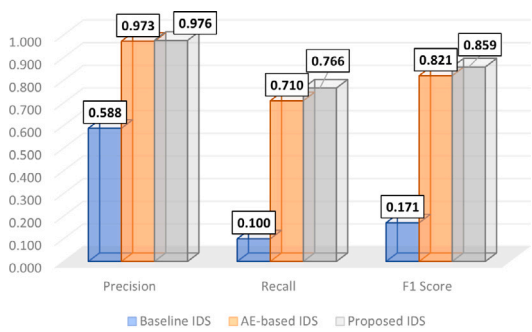

Fig. 9. Ablation study results under the ZOO attack, comparing the baseline classifier (undefended), AE-based classifier (partially defended), and the proposed RADIANT IDS.

defense, demonstrated a significant enhancement in performance, reinforcing the importance of reactive defensive mechanisms. Furthermore, when compared with state-of-the-art methods (Table 6), the proposed RADIANT IDS achieved the highest F1 score of 85.9% and smallest FPR of 1.4%, showcasing its ability to maintain a balanced performance across metrics. While the detection mechanism proposed by Debicha et al. (2021) and Pawlicki et al. (2020) surpassed RADIANT IDS in either precision or recall, their lower F1 scores indicate a trade-off between these metrics, underscoring the strength of RADIANT IDS in achieving overall consistency. This balance is particularly significant, as RADIANT IDS accomplishes such competitive performance without requiring adversarial retraining, highlighting its practicality and efficiency in addressing adversarial threats. These findings reaffirm the effectiveness and robustness of the proposed IDS in defending against ZOO attacks.

8. Conclusion

Machine learning-based IDSs have become indispensable tools for safeguarding ICS systems against cyberattacks due to their effectiveness and adaptability. However, these systems are susceptible to adversarial attacks, commonly referred to as AML attacks, which can compromise their accuracy and reliability. In the context of ICS infrastructures, these attacks can have severe consequences, as malicious actors may manipulate data points to evade detection, leading to delayed threat identification and potentially extensive damage. Therefore, it is imperative to fortify IDSs against adversarial attacks, whether through proactive approaches or reactive defense mechanisms.

This paper investigates the vulnerabilities of finely tuned classifiers when subjected to two distinct adversarial threats, simulated under a realistic threat model. Furthermore, it introduces an innovative reactive defense mechanism, RADIANT IDS, designed to mitigate adversarial threats without relying on prior adversarial training. To assess the effectiveness of the proposed defense mechanism, we conducted a comprehensive evaluation using a widely recognized power system dataset and a robust evaluation framework.

Our evaluation compared the performance of RADIANT IDS with baseline classifiers and state-of-the-art adversarial defense mechanisms across three distinct scenarios. In the first scenario, which involved unperturbed test set data, all classifiers, including the proposed IDS, exhibited similar performance. However, under the subsequent adversarial scenarios of HopSkipJump attack and the ZOO attack, RADIANT IDS consistently outperformed both baseline and state-of-the-art classifiers without any adversarial training.

Beyond adversarial robustness, the RADIANT module is designed to be highly adaptable, allowing seamless integration with any IDS while adjusting to the computational constraints of the deployment system. Unlike traditional adversarial defenses that require retraining with adversarial samples, which is computationally expensive and often degrades performance on benign traffic, RADIANT IDS operates as an independent adaptive layer. This ensures that the IDS remains effective across both normal and adversarial conditions without the need for continuous retraining. Furthermore, its design leverages parallelizable feature extraction techniques, ensuring real-time operability in industrial control networks.

These findings highlight the critical vulnerability of finely tuned classifiers to adversarial threats and underscore the necessity of robust defense mechanisms in ICS contexts. The results also emphasize the importance of exploring innovative reactive defenses, such as RADIANT IDS, as a complement to traditional adversarial training-based approaches, offering a practical and effective solution for enhancing adversarial robustness.

9. Future work

In this study, we evaluated the proposed reactive defense mechanism under the context of grey-box attacks and two distinct adversarial

threats. However, a key limitation of the RADIANT IDS is its potential vulnerability to white-box attacks, where an attacker possesses detailed knowledge of the preprocessing modules, including the defense mechanism itself. In such scenarios, attackers can craft highly tailored adversarial samples that bypass the defense, as the reactive mechanism's inner workings are fully exposed. While this represents a limitation, it is important to note that white-box attacks are relatively rare in practical applications, as black-box and grey-box attacks are far more common in real-world threat models. In these more prevalent attack scenarios, our approach provides adequate protection and maintains robust performance.

As part of our future work, we plan to expand our threat model to include white-box attack scenarios, where attackers exploit knowledge of the deployed intrusion detection model. To counter this, we will introduce ensemble meta-classifiers, preventing attackers from developing surrogate models and evading detection. These classifiers will operate in parallel, continuously assessing the prediction credibility of the main classifier, with detected deviations dynamically altering the processing pipeline. Since the decision boundaries of multiple classifiers differ, attackers will find it difficult to craft adversarial samples that generalize across all classifiers. Additionally, we will incorporate stochastic preprocessing techniques and adaptive modules to introduce unpredictability, further preventing attackers from consistently evading detection. By integrating these adaptive defenses with our reactive module, we aim to enhance IDS robustness against both black-box and white-box attacks. This holistic approach is expected to offer more robust protection against the dynamic and evolving threat landscape in the domain of ICS security.

CRediT authorship contribution statement

Irfan Khan: Writing – review & editing, Methodology, Formal analysis, Conceptualization. **Syed Wali:** Writing – review & editing, Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Yasir Ali Farrukh:** Writing – review & editing, Writing – original draft, Validation, Methodology, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This work relates to Department of Navy award (N00014-23-1-2626) issued by the Office of Naval Research. Any opinions, findings, and conclusions or recommendations expressed in this material are those of the author(s) and do not necessarily reflect the views of the Office of Naval Research.

Data availability

Data will be made available on request.

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